LOVJYOT SINGH

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PROFESSIONAL SUMMARY

CSE graduate (USICT, GGSIPU, 2026). Shipped three live products on **React**, **Next.js**, **TypeScript**, **Node.js**, **PostgreSQL**, and **MongoDB**: a serverless real-time collaborative editor, an AI mock-interview platform with rubric grading, and a spreadsheet engine built from scratch. Immediate joiner. Portfolio: lovjyotsingh.github.io.

EDUCATION

Bachelor of Technology – Computer Science and Engineering
USICT, Guru Gobind Singh Indraprastha University, New Delhi — July 2022 – July 2026

PROJECTS

SyncFlow – *Real-Time Collaborative Workspace* — Live Code

- **Serverless sync engine:** Built a multiplayer block editor in **Next.js 16**, **TypeScript**, and **BlockNote** on **Vercel**, synced by a custom **Yjs** provider over HTTP with one **Postgres** transaction per round trip.
- **No lost edits:** Append-only update log on **Neon Postgres** with a settle window so concurrent edits are never skipped, plus background compaction into snapshots; tested with **Vitest** against a real Postgres.
- **Sharing:** Live cursors and presence, rotatable share links, email invites, and file sharing on **Vercel Blob**.

OfferForge AI – *Full-Stack AI Mock Interview Platform* — Live Code

- **Structured loops:** 7 role tracks as fixed sequences of sections: DSA, system design, OOP, SQL, behavioral.
- **Rubric grading:** Designed a **Gemini** and **OpenRouter** pipeline that scores each answer on a 4-criterion rubric and returns strengths, gaps, a model answer, and a hire call, with a curated question bank as fallback.
- **Session integrity:** JWT auth on **MongoDB**, a server-held question flow with atomic submit and skip, and a score-trend dashboard; **React/Vite** client and **Express** API on Vercel.

Gridline – *Spreadsheet Engine Built from Scratch* — Live Code

- **Formula engine:** Wrote a tokenizer and Pratt parser in **TypeScript** with Excel's operator precedence, 22 functions, and errors such as #DIV/0! that propagate to dependent cells; covered by **Vitest** tests.
- **Dependency graph:** Recalculates only the cells downstream of an edit, in topological order (Kahn's algorithm), and flags cycles; one change in a 10,000-row column recalculates 2 cells, not 10,000.
- **Virtualized grid:** Draws only the ∼100 visible cells of a 10,000-row sheet; Excel-style copy/paste, undo.

TECHNICAL SKILLS

Languages	Java, TypeScript, JavaScript, SQL, HTML, CSS
Frameworks	React.js, Next.js 16, Node.js, Express.js, Socket.io, Yjs, Tailwind CSS, Vite
Databases & Cloud	PostgreSQL (Neon), MongoDB, MySQL, Redis, Docker, Vercel, Vercel Blob
Tools & DevOps	REST APIs, Serverless, Microservices, JWT, Webhooks, CI/CD, Vitest, Git, Postman
Fundamentals	DSA (200+ problems solved), System Design (HLD/LLD), DBMS, OS, Networks

PROGRAMS & CERTIFICATIONS

IBM Web Development Program – *Front-End Engineering Track* — *July 2025 – August 2025*

- Project-based training in **React.js**, **TypeScript**, and **Tailwind CSS**: components, state, and accessibility.